

Testing Requirement Analysis Method Based on Signal Path for High Reliability Software

Yunyun Ma, Jinbo Wang, Yuelong Qu, Tao Zhang

Technology and Engineering Center for Space Utilization

Chinese Academy of Sciences

Beijing, China

{myy|wangjinbo|qyl|zt}@csu.ac.cn

Abstract—In high reliability embedded software testing, the software requirements are often described and partitioned from a view of design module, which often lead to inadequate test requirement analysis and undetected defects. In order to solve this problem, a test requirement analysis method is proposed. In this method, the signal paths of the tested software on configuration item (CSCI) level are analyzed to reconstruct the combination relationship between functions. Then, test items are generated by performing multidimensional decomposition on test meta-items. Finally, a combination of multivariate test methods is used to generate the final testing requirement specification. The validity and feasibility of this method have been fully verified in a number of aerospace software test projects.

Keywords—testing requirement; high reliability software

I. TECHNICAL SCHEME OF TESTING REQUIREMENT ANALYSIS METHOD BASE ON SIGNAL PATH

A practical problem in high reliability embedded software field is that the software requirement specification of the tested software configuration items are often described and partitioned from a view of design module^[1]. In this case, testing requirements analysis performed on functional points explicitly defined in the tested software requirements can easily lead to the problems such as unidentified hidden requirements, incomplete observation content, etc. In order to solve these problems, a testing requirement analysis method based on the signal path is proposed in this paper.

The technical framework of this testing requirement analysis method consists of four layers: independent function point, test business, test meta-item and test item.

The testing requirement analysis based on signal path is a process of continuous iteration and improvement: 1) Sort out the function points from software requirements. 2) Generate test business by combining the discrete function points along the signal paths running through the CSCI. 3) Determine the meta-item and perform multidimensional decomposition to generate independent testing items. 4) Analyze every test item using multivariate methods to get test adequacy requirements, including test scenario, key point of test content, etc. 5) Communicate and confirm the defects with related parties. 6) Redo above steps based on the confirmation if necessary.

The framework and implementation mechanism of the testing requirement analysis method based on signal path method is shown in Fig. 3.

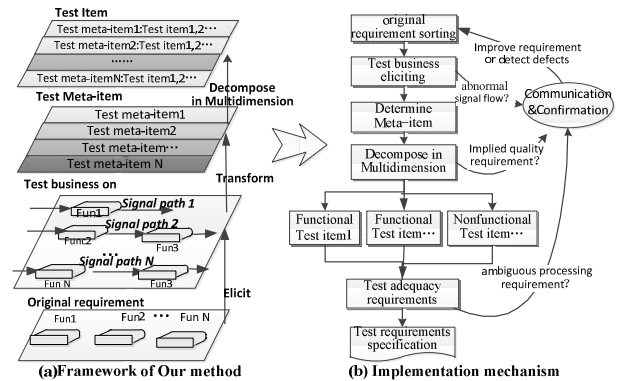


Fig. 3. Test requirement analysis method based on signal path

Take 3 practical aerospace software test projects as example. All these 3 software have been tested for several rounds by other testing team using common test requirement analysis method, and all the bugs have been fixed before test. After using our method, 175 new defects are detected, and the newly detected defect rate has reached up to 5.83 per thousand code lines, which is still very high to high reliable software. Therefore, our method is proved to be effective to improve the defect detection ability to high reliable software.

TABLE I. APPLICATION EFFECT IN PRACTICAL TEST PROJECTS

Test Object	LOC	Meta-item	Testing Item		Test Case	Defect	Defect Rate
			Functional	Other quality			
A	7000	20	45	32	580	41	5.86‰
B	3000	12	22	23	303	45	15.00‰
C	20000	11	18	32	895	89	4.45‰
Total	30000	33	85	87	1778	175	5.83‰

II. CONCLUSION

This paper presents a new testing requirement analysis method based on signal path. Practice proved that the method has good application effect. In the future we plan to do further study on test case automatic generation based on this method.

REFERENCES

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